

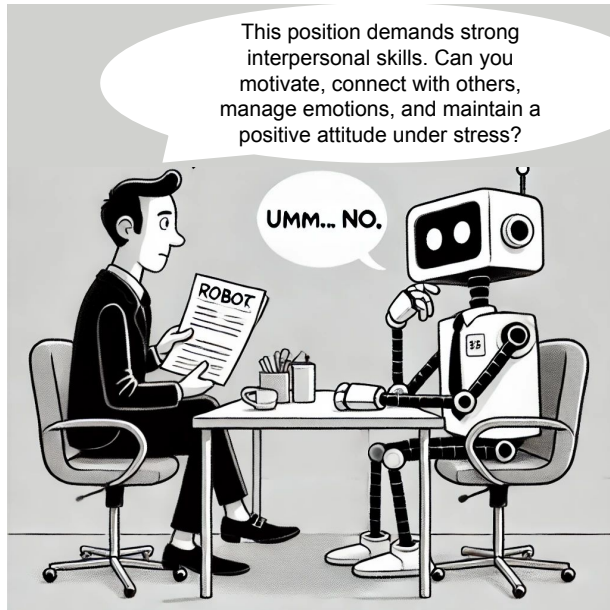
Week 6: Business applications of LLMs

Course: BUSM2725 AI and Analytics for Business Applications

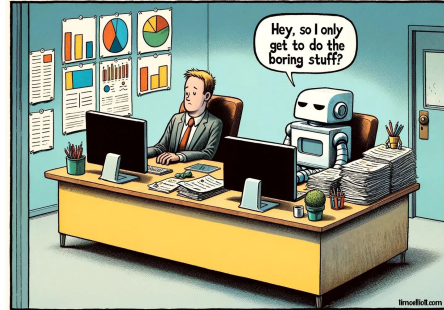
Outline

- Common use cases of GenAI in Business
- Activity: Explore AI Use Cases from Microsoft
- Activity: Document Understanding and hands-on activities with SPARK
- Activity: Computer Vision tasks with SPARK
- Activity: Basic Data Analysis with SPARK
- Activity: Research analysis workflow with SPARK
- Demo: Power Automate and GPT example.

AI in Workplace



Source: DALL-E, Captioned by Su Nguyen



Source:

<https://timoelliott.com/blog/cartoons/artificial-intelligence-cartoons>



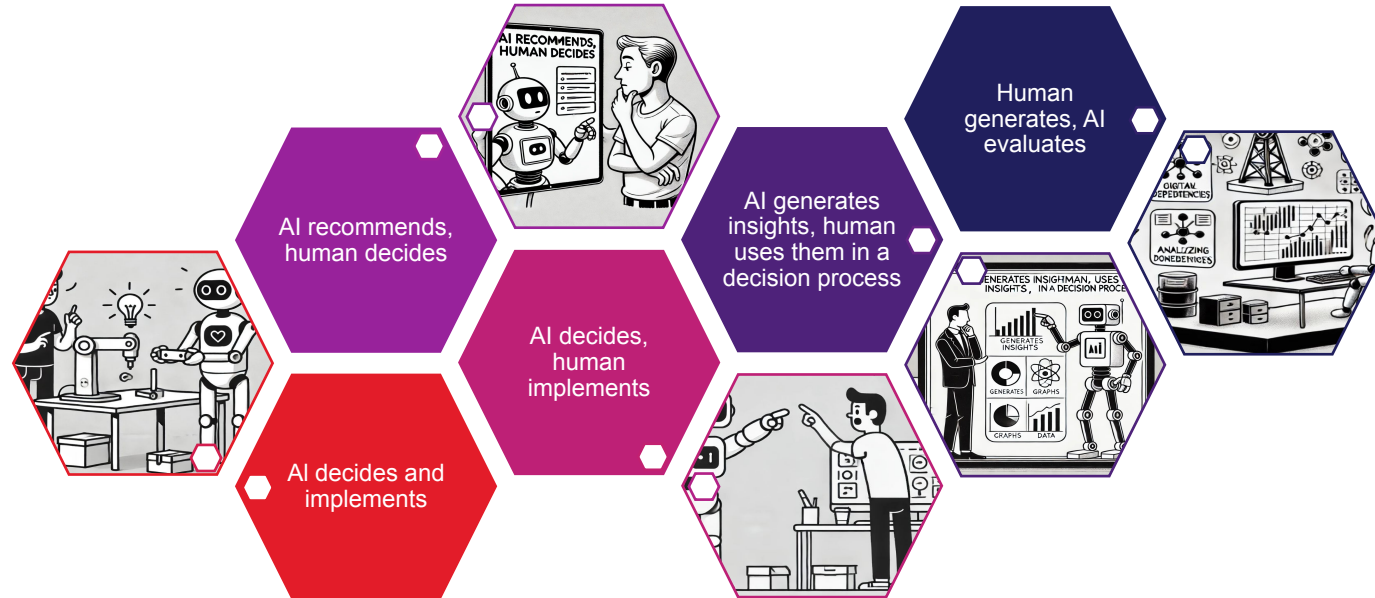
Source: DALL-E,
Prompt by Su Nguyen

AI Fast Facts	Percentage of desk workers who have used AI for work	Percentage growth in desk workers using AI since September 2023	Percentage of AI users who are seeing productivity gains	Percentage of desk workers who consider AI-generated outputs completely trustworthy
	32%	60%	82%	7%
	37%	35%	79%	5%
	31%	56%	81%	13%
	36%	35%	86%	9%
	20%	7%	69%	1%
	32%	66%	85%	6%
	32%	79%	82%	7%

slack
Slack survey conducted January 10, 2024 - January 29, 2024. Number of completed responses = 10,281.

“Despite excitement about AI, Australian desk workers lack trust, with only 5% fully trusting AI to handle tasks, which is below the global average. Additionally, only 13% feel they have adequate education and training to use AI effectively.”

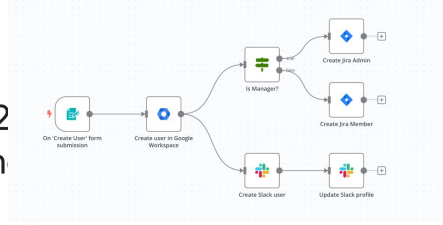
Better Together: How Humans and AI Interact?



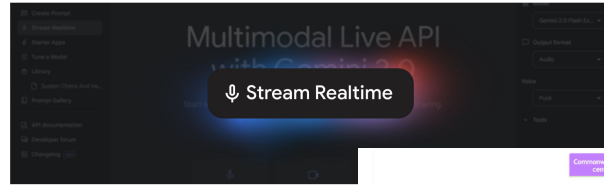
Tools are there, but the choice is tough ...

Gemini 2 Real-Tim

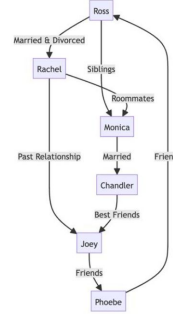
DEC 23, 2024
Ivan Solovyev
Product Manager
Shreshtha Basu Mallick
Group Product Manager
Gemini API



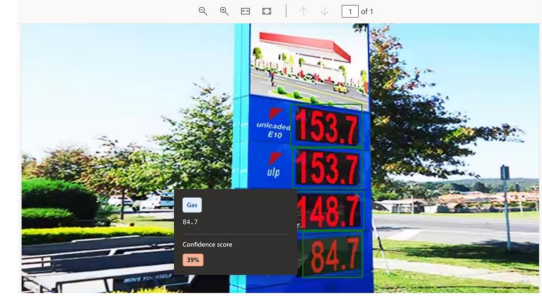
Share



Val (GPT-4o) 6:28 PM
Sure! Here's a Mermaid diagram showing the relationships between the main characters in Friends:



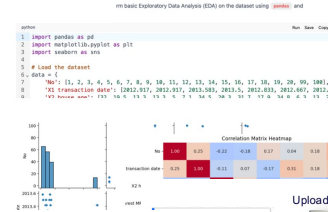
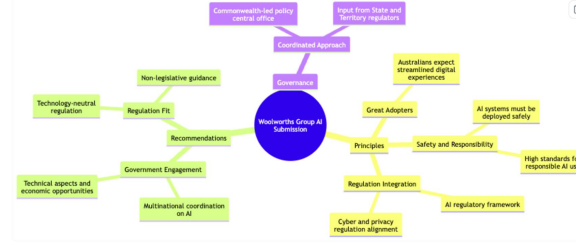
Quick test



Learn how you can improve your model Start over Close

ode to conduct EDA on the data table above

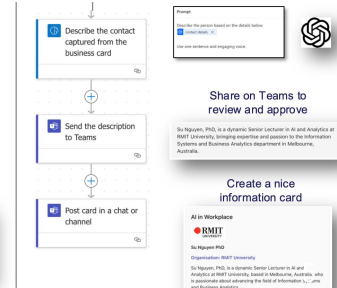
This diagram outlines some of the key relationships between the characters.



 **Hugging Face**



Amazon SageMaker



Val (GPT-4o)

RMIT's Virtual Assistant for Learning AI (OpenAI GPT-4o).

Suggested

Explain how to structure a convincing argument
Prompt

What is responsible use of AI?

+ Send a Message

NotebookLM



Potential Outcomes Achieved with AI



Enriching Employee Experiences

Streamlining tasks for creative work

Reinventing Customer Engagement

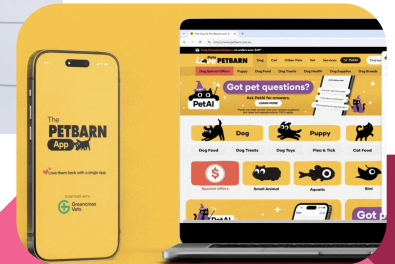
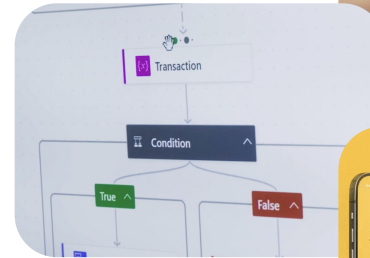
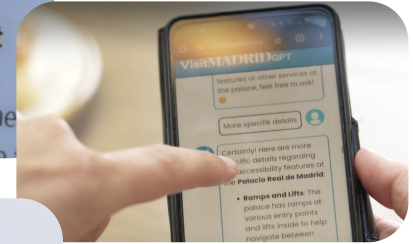
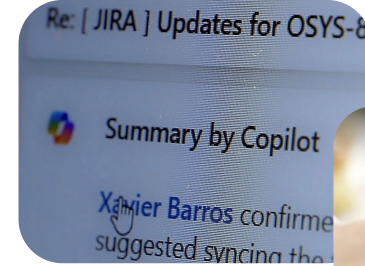
Personalizing customer interactions

Reshaping Business Processes

Reimagining processes for growth

Bending the Curve on Innovation

Accelerating innovation and market entry



Source: [How real-world businesses are transforming with AI](#), Microsoft

AI's long tail (customization) problem

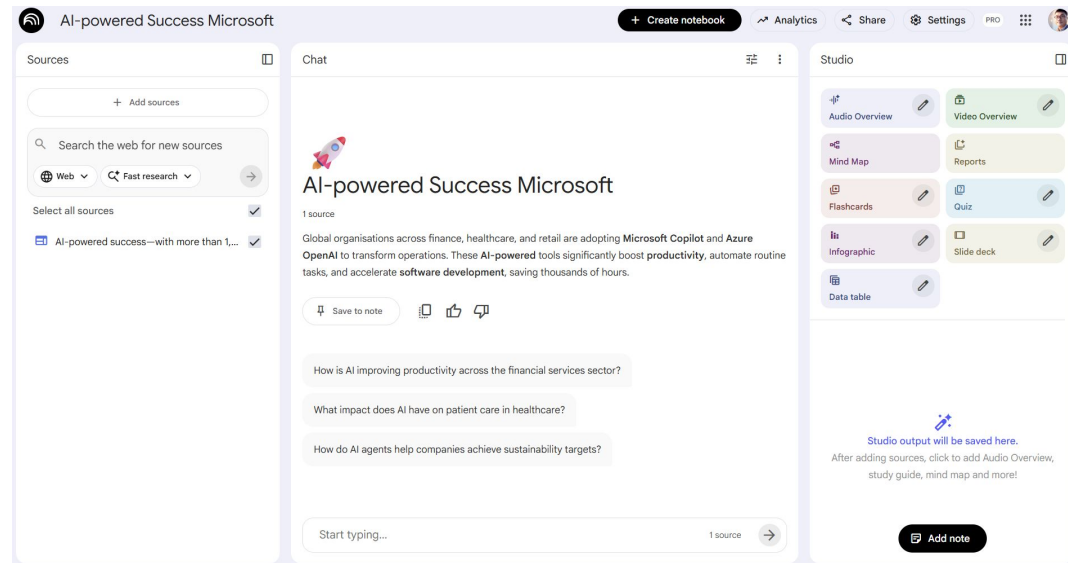


Source: [How AI Could Empower Any Business | Andrew Ng | TED](#)

All potential AI projects, sorted in decreasing order of value

Activity: Explore AI Use Cases from Microsoft

- Visit AI-powered success cases from Microsoft [here](https://notebooklm.google/)
- Have a quick look and copy the link
- Visit <https://notebooklm.google/>
- Access notebooklm tool
- Create new Notebook
- Click Website and paste the link above – click Insert

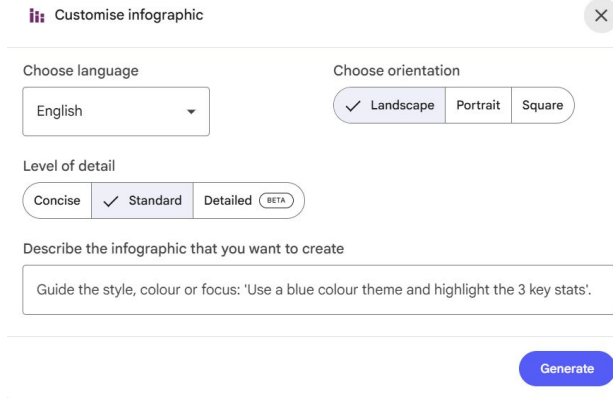
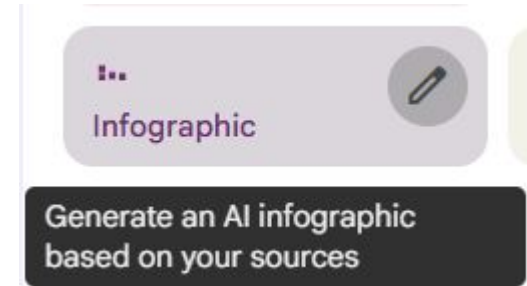


Explore using Mind Map



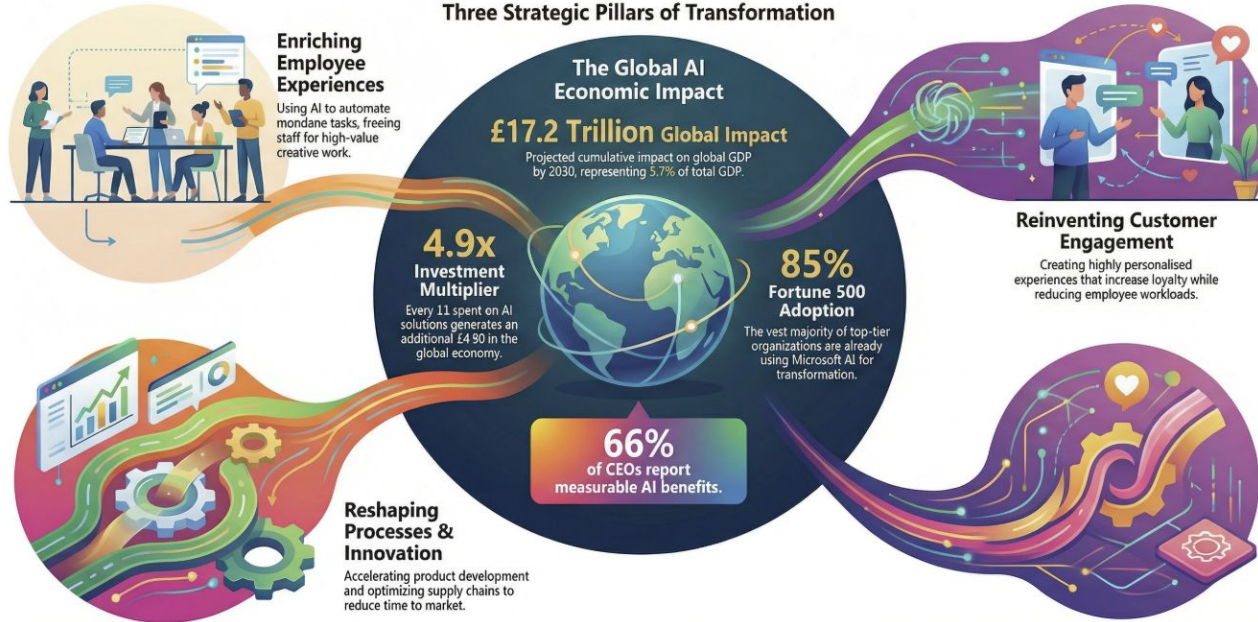
Generate Infographics

- Click Infographic in Studio pane
- Click Edit
- Describe the infographic you want to create, think of the following:
 - Who you would like to present your infographic?
 - Topics to present (process automation)
 - Level of details
 - Layout

A screenshot of the 'Customise infographic' dialog box. It has a title bar with a close button. The dialog contains several settings: 'Choose language' with a dropdown menu set to 'English'; 'Choose orientation' with three buttons: 'Landscape' (checked), 'Portrait', and 'Square'; 'Level of detail' with three buttons: 'Concise', 'Standard' (checked), and 'Detailed' (with a 'BETA' badge); and a text area for 'Describe the infographic that you want to create' containing the text 'Guide the style, colour or focus: 'Use a blue colour theme and highlight the 3 key stats''. At the bottom right is a blue 'Generate' button. A large pink arrow points from the 'Generate an AI infographic' button in the previous image to this dialog box.

AI-Powered Success: Driving the Global Transformation

Three Strategic Pillars of Transformation



© NotebookLM

Example of General Infographic

Example of Customised Infographic



Customise infographic

Choose language

Tiếng Việt

Choose orientation



Landscape

Portrait

Square

Level of detail

Concise

✓ Standard

Detailed

BETA

Describe the infographic that you want to create

Create an infographic focusing on how AI is used to automate operations in manufacturing industry. Arrange the infographic as three layouts (top to bottom), first layer focuses on AI techs, second layer focuses on tasks, and the last layer focuses on expected outcomes.

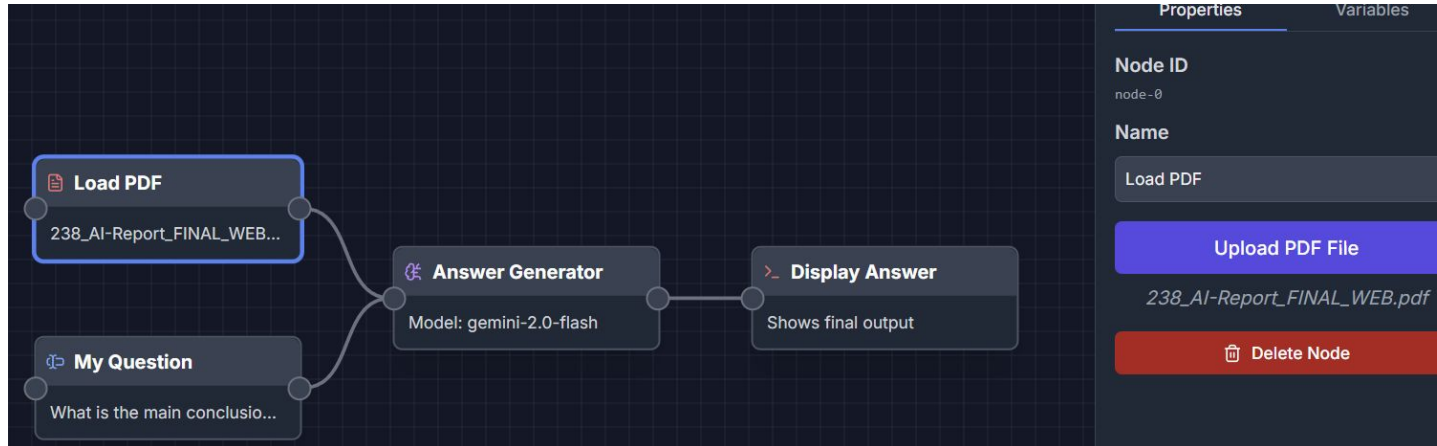
Generate

Activity: Document Understanding and hands-on activities with SPARK

- Review how SPARK handle input files
- Analyse submission letters to AASB
- Build a flow to analyse these letter
- How to handle multiple files

Review last week activity

Load the first example in Gallery



Upload the "Accelerating Australia's AI Agenda" report in the Load PDF node.

▶ Run

Agent Output

The main conclusion of this document is that Australia's embrace of Artificial Intelligence (AI) presents an **immediate and transformative opportunity** crucial for addressing the nation's productivity challenge. The document stresses that the decisions and foundational investments made now in developing, deploying, and regulating AI will critically determine Australia's **long-term economic prosperity and global competitiveness** for the next two decades and beyond, deciding whether the nation gains an enduring competitive edge or faces decades of playing catch-up.

Analyse submission letters to AASB

The AASB's Exposure Draft **ED SR1** *Australian Sustainability Reporting Standards – Disclosure of Climate-related Financial Information* which proposes climate-related financial disclosure requirements closed for comment on Friday, 1 March 2024. Stakeholders strongly encouraged greater alignment with the International Sustainability Standards Board's IFRS S1 & IFRS S2 baseline. Our **Climate Disclosures Consultation Update** explains the key changes following the consultation, including the areas where the AASB has taken a back to baseline approach.

ED SR1 includes three draft Australian Sustainability Reporting Standards (ASRS Standards):

1. [draft] ASRS 1 *General Requirements for Disclosure of Climate-related Financial Information*, developed using IFRS S1 *General Requirements for Disclosure of Sustainability-related Financial Information* as the baseline but with a scope limitation to climate-related financial disclosure;
2. [draft] ASRS 2 *Climate-related Financial Disclosures*, developed using IFRS S2 *Climate-related Disclosures* as the baseline; and
3. [draft] ASRS 101 *References in Australian Sustainability Reporting Standards*, developed as a service standard that would be updated periodically to list the relevant versions of any non-legislative documents published in Australia and foreign documents that are referenced in ASRS Standards.

Stakeholders were invited to provide feedback on ED SR1 by submitting a comment letter, completing an online survey or attending a roundtable discussion. The comment letters are found [here](#).

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Stakeholders were invited to provide feedback on ED SR1 by submitting a comment letter, completing an online survey or attending a roundtable discussion. The comment letters are published below.

Submission received from:
Central Coast Council
ADCO Construction
Insurance Council of Australia (ICA)
ABS
T. Rowe Price
Confidential
Deakin University IR Centre
ASIC

Download 2-4 letters of interests (e.g. ABS, UniSuper, Deloitte)

In the demonstration here, I will use letters from banking industry

Create the flow to analyse submission letters

- Use PDF file as the input node
- Edit the prompt to decide how the document will be analysed
- Upload the submission file in the PDF file node
- Click Run to start analysing



Agent Output

National Australia Bank (NAB) is **strongly supportive** of the Exposure Draft's objective to create an environment that promotes consistent, comparable, and reliable climate-related financial disclosures in Australia. They believe these proposals are **overall in the best interests of the Australian economy**. However, NAB emphasizes the critical need for **close alignment with equivalent international reporting standards** (IFRS Sustainability Disclosure Standards) to ease implementation for multi-jurisdictional entities and prevent Australia from falling behind global developments.

Reflection

- Review the letter and the output from SPARK
- Is the AI output reliable?
- How many input tokens are used? And how many output tokens are generated?
-

Handle Multiple Files

- Replace the PDF File node with Iterator node
- In Iteration Type, select File List
- Click Upload Multiple Files (might need to unselect the node and select it again to refresh the property window of this node)
- Remember to change the node referred in the prompt

Properties

Variables

Node ID

node-3

Name

Iterator

Iteration Type

File List

Upload Multiple Files

edsr1_aba.pdf, edsr1_coba.pdf, edsr1_nab

Delete Node

Execution Outputs

Batch Output #2

3 iterations from "Iterator" 03/02/2025, 14:31:34

Input: edsr1_aba.pdf

The Australian Banking Association (ABA) generally supports the intent of the Exposure Draft ED SRI Australian Sustainability Reporting Standards but strongly advocates for maximal alignment with international standards, particularly IFRS S1 and S2, to ensure global comparability and future-proofing.

Key points of their sentiment include:

Strong Preference for International Alignment: They recommend adopting Option 2 for presenting core content across two ABB standards to maximise alignment with international benchmarks and facilitate future updates. They also suggest developing mapping documentation to bridge global and domestic reporting classifications (e.g., ANZSIC vs. global codes).

Opposition to Superannuation Exclusion: The ABA strongly opposes excluding large superannuation entities, arguing that the cited challenges (e.g., Scope 3 complexity) are generic to the financial sector and that exclusion would undermine legislative intent and public confidence.

Concerns with Data & Clarity: They highlight significant challenges with data quality, availability, and measurement methodologies, especially concerning Scope 3 GHG emissions, requesting more flexibility on data age and clearer definitions (e.g., "reasonable and supportable information"). They also seek greater clarity on various aspects, including the intended location of disclosures, interim reporting, and the specific application of certain metrics.

Auditing & Assurance Challenges: The ABA acknowledges the complexities and nascent state of global standards for sustainability assurance, particularly for forward-looking statements and Scope 3 emissions, suggesting that limited assurance may often be more appropriate than reasonable assurance.

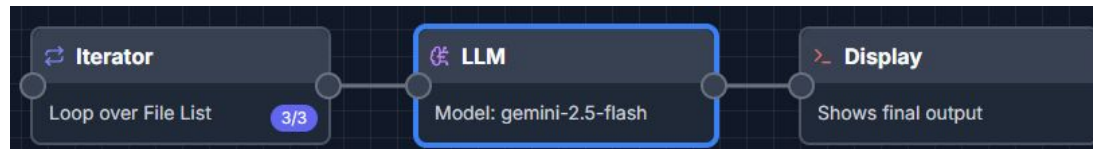
Overall Benefits: Despite the challenges and requests for modification, the ABA believes that ISSB-aligned disclosure standards will result in useful information for users and are in the best interests of the Australian economy by attracting capital, managing risks, and fostering a clear transition to net-zero. They also stress the importance of complementary government work on national climate data infrastructure.

Input: edsr1_coba.pdf

COBA generally welcomes the AASB's draft Australian Sustainability Reporting Standards and the opportunity to comment. The organisation expresses strong support for the full adoption of the ISSB's IFRS S2 and S1 standards in Australia, with modifications limited to those necessary for local context, to ensure international alignment and streamline reporting.

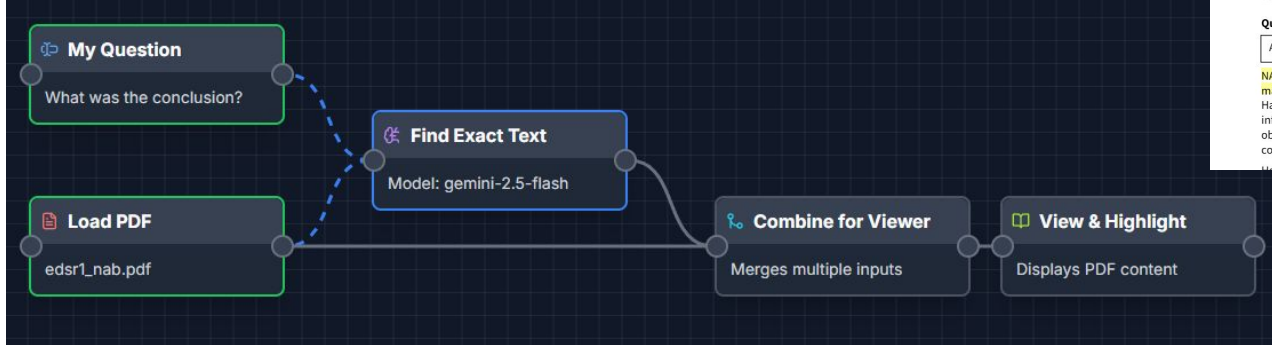
However, COBA articulates several key concerns and recommendations:

Commencement Date: Strongly recommends delaying the commencement date from 1 July 2024 to 1 January 2025 to



Text Highlighter

- Check the PDF Text Highlighter example in Gallery
- Click the View & Highlight node to view pdf file with texts (extracted by LLM) highlighted.



Another challenge is that an entity's sustainability report is expected to include both disclosures required under ASRS Standards as well as those made under voluntary frameworks which may require different levels of assurance.

NAB is supportive of Treasury's proposal for assurance requirements to be phased in over time. However, during the phase-in period, it may be difficult for the users of the sustainability report to distinguish between disclosures that have been subject to differing levels of assurance.

Question 33 – Information usefulness

Would the proposals result overall in climate-related financial information that is useful to users?

NAB believes the proposals will result overall in climate-related financial information that is useful to users of the sustainability report, including in relation to physical risks arising from climate change, risks associated with the transition to a low carbon economy, as well as climate-related opportunities. NAB acknowledges that entities are at different stages of reporting maturity and that climate science continues to evolve. NAB expects that the best practice of sustainability reporting will continue to develop over time to better meet users' information needs.

Question 34 – Best interests of the Australian economy

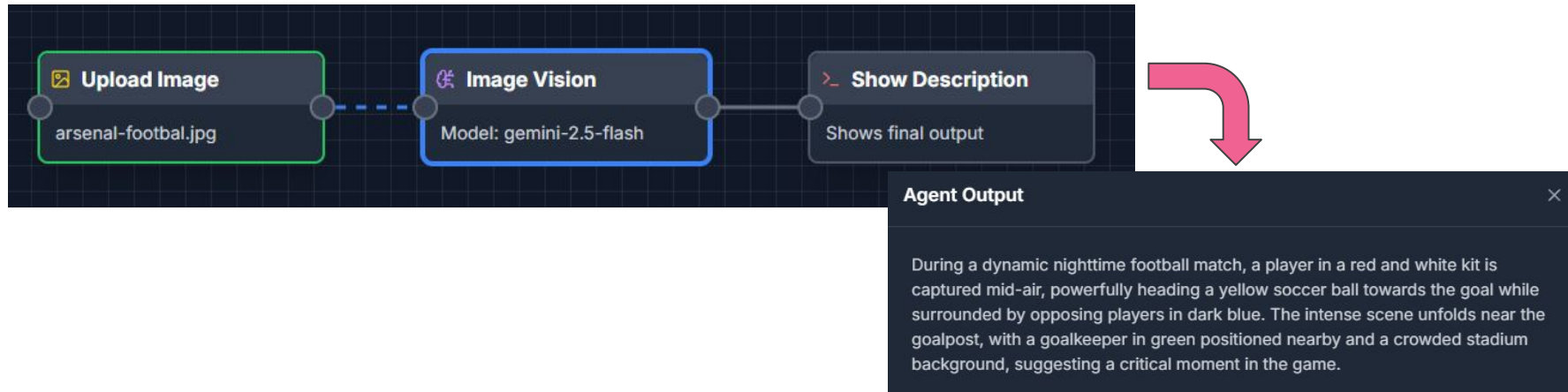
Are the proposals in the best interests of the Australian economy?

NAB believes that proposals to enhance transparency and comparability of information on how entities are managing climate-related risks and opportunities are overall in the best interests of the Australian economy. Having an improved understanding of emissions and climate-related risks and opportunities should enable informed action that can help Australia meet its international UN Framework Convention on Climate Change obligations; mitigate, adapt and build resilience to climate risk; and maintain quality of life and competitiveness in the global trade environment.

However, it is important that the introduction of ASRS Standards is assessed with consideration for the

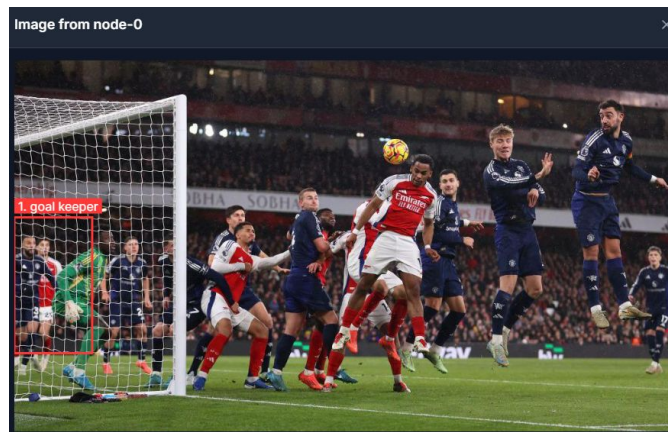
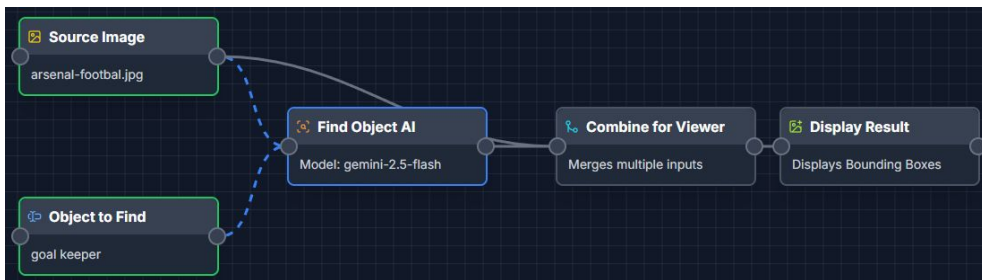
Activity: Computer Vision tasks with SPARK

- Select the Image Describer example
- Upload the arsenal-football.jpg in the first node
- In the Image Vision (LLM node), use this prompt: “describe the image in two sentences.”

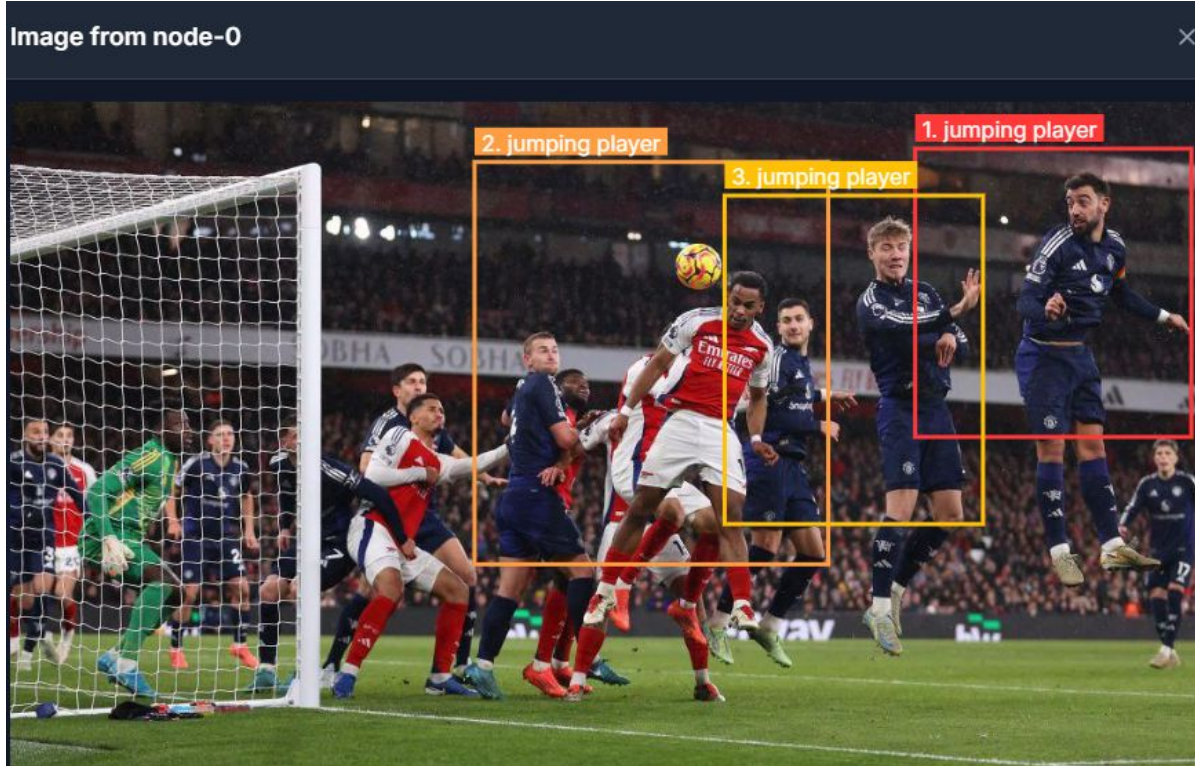


Object Detection

- Select “Specific Object Detection” example in the gallery
- Upload the image earlier and change the description of object to find



Multiple objects and important notes



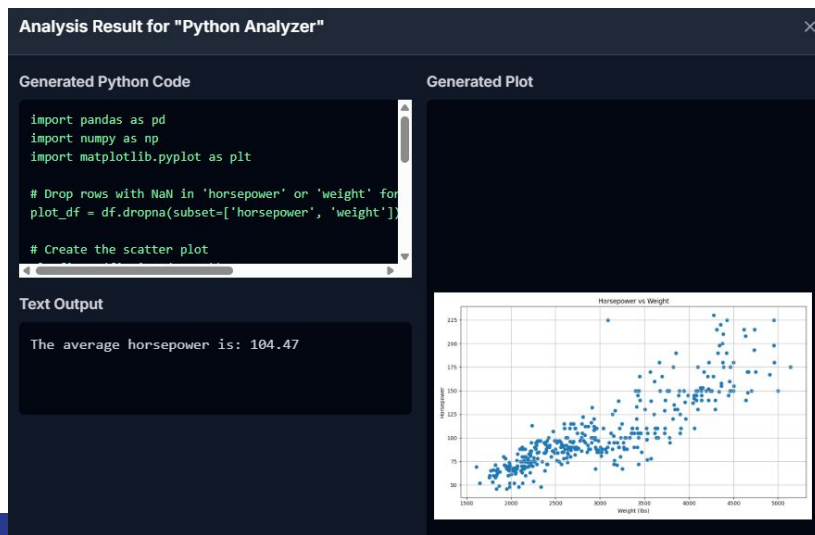
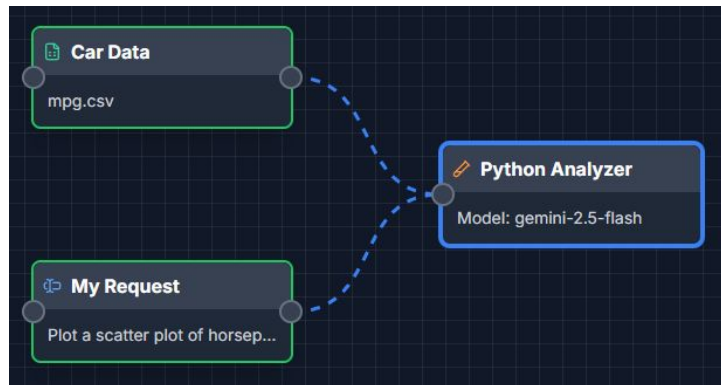
Note: object detection is a bit unstable in the current SPARK version due to the updated APIs.

Object detection quality depends on the selected AI model.

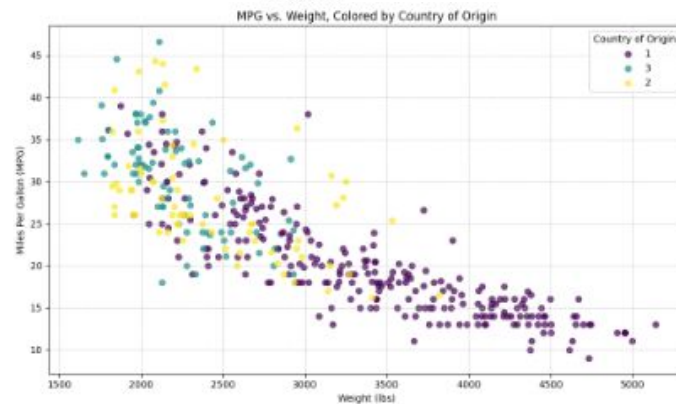
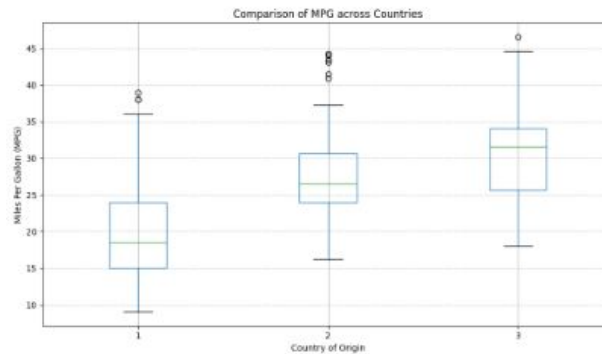
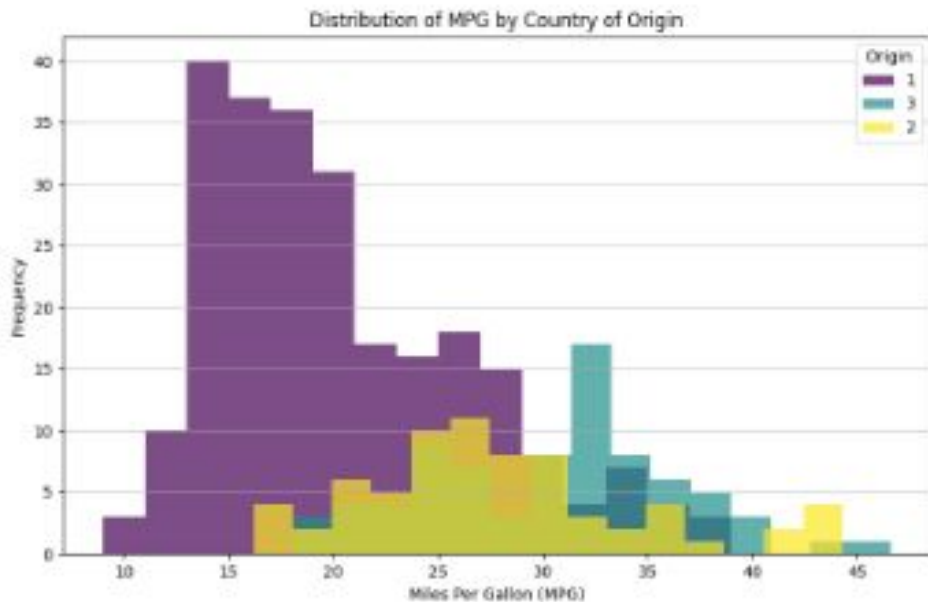


Activity: Basic Data Analysis with SPARK

- Select the CV Analysis with Python example
- Upload the mpg.csv file to the csv node
- Use the existing prompt: “Plot a scatter plot of horsepower vs weight. Also, print the average horsepower.”

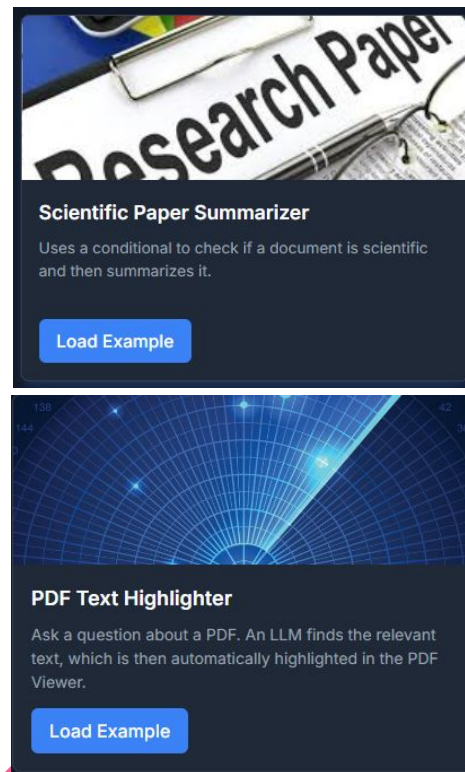


Try to edit the prompt to generate the charts below



Activity: Research analysis workflow with SPARK

- Use the “automated-formulation.pdf” file for these two examples.
- Try to get the following insights:
 - The motivation for this paper
 - The main methodology used in the paper
 - What are the main outcomes and conclusions
 - See the description of proposed techniques in the pdf file

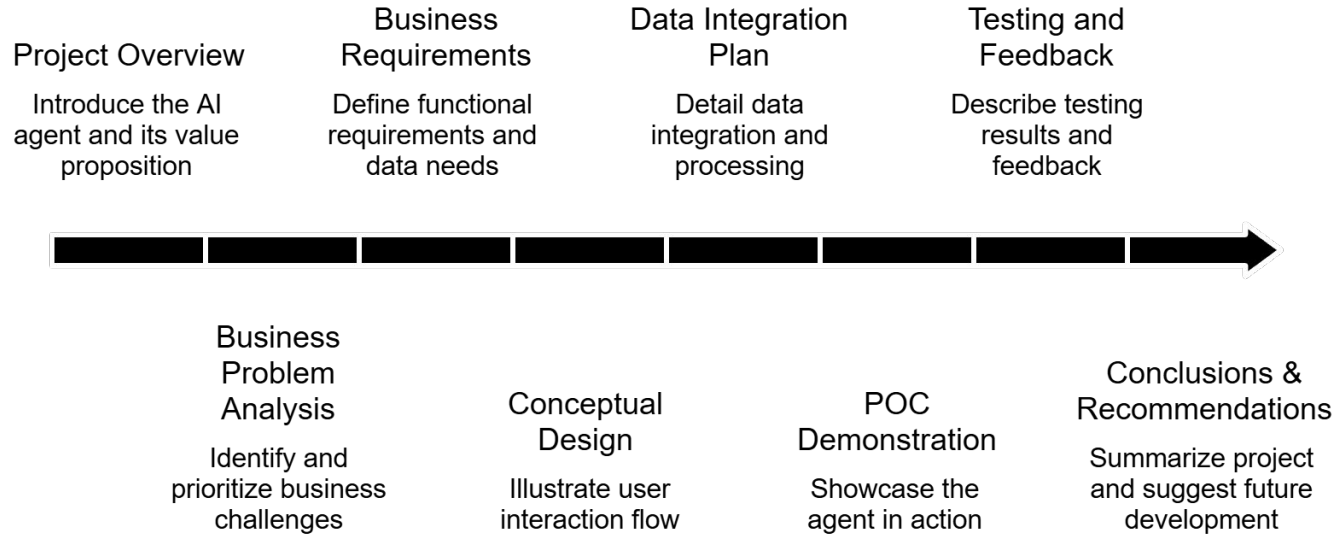


Reflection

- Based on the hands-on activities so far, are you able to identify what AI technologies used by businesses in the AI-powered success stories (from Microsoft)?
- Please review the Statistics tab and think about how much computing units are used in your activities? Are your consumption patterns similar to your peers? If different, why?



Q&A for Assessment 2



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